

Student Guide — middle / module-09-subtraction-nikhilam

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — Module 9 Nikhilam Subtraction (Middle)

Name: _____ Class: _____ Section: _____

This is your workbook for the next two weeks. Bring it to every class.

The Sūtra

Nikhilam Navataścaramaṁ Daśataḥ

“All from nine and the last from ten.”

— Tirthaji, *Vedic Mathematics* (1965), sūtra #2

Day 1 — The Sūtra and the Hook

My baseline time (standard borrowing) for $10\ 000 - 4567$: _____ seconds

My first try with Nikhilam: _____ seconds

I will recite the sūtra from memory by Day _____.

Day 2 — Complement to 100

The rule for 2-digit numbers AB:

- Tens digit: $9 - A$
- Units digit: $10 - B$

Practice (write the complement):

73 → ___ 41 → ___ 28 → ___ 86 → ___ 95 → ___

17 → ___ 60 → ___ 52 → ___ 34 → ___ 89 → ___

Speed check: 20 complements in under 60 seconds [Y / N]

Day 3 — Complement to 1000

The rule for 3-digit numbers ABC:

- First digit: **9 - A**
- Second digit: **9 - B**
- Last digit: **10 - C**

Practice:

738 → ___ 419 → ___ 256 → ___ 803 → ___ 999 → ___

And to 10 000: 2 347 → ___ 4 567 → ___ 8 888 → ___

Day 4 — Subtracting from a Power of Ten

Padding rule: if the subtrahend has fewer digits than the base, pad with leading zeros.

Practice (write the answer):

1000 - 358 = ___ 10 000 - 6 427 = ___ 1000 - 72 = ___

100 000 - 12 345 = ___ 10 000 - 9 = ___

Day 5 — Mid-module Retest

Day 1 baseline: ___ sec → Day 5 Nikhilam: ___ sec → **% change:** ___

My project choice: _____ (chart / video / poster) **Partner:** _____

Day 6 — Why It Works

The key identity:

$$10^n - A = (10^n - 1) - A + 1$$

In your own words (paragraph):

Day 7 — Two's Complement

	Base 10 (Nikhilam)	Base 2 (Two's Complement)
Maximum digit	9	1
"All from max"	All from 9	Flip each bit
"Last from base"	Last from 10	Add 1

Practice: $7 - 5$ in 4 bits.

$7 = \underline{\hspace{1cm}}$ $5 = \underline{\hspace{1cm}}$ two's-comp of 5 = $\underline{\hspace{1cm}}$ Sum = $\underline{\hspace{1cm}}$ Drop top bit $\rightarrow \underline{\hspace{1cm}}$

Answer: $\underline{\hspace{1cm}}$

Day 8 — General Subtraction

The identity:

$$\mathbf{M} - \mathbf{S} = (\mathbf{M} - \mathbf{B}) + (\mathbf{B} - \mathbf{S}), \text{ where } \mathbf{B} \text{ is a power of ten with } \mathbf{B} > \mathbf{S}.$$

Practice:

$8\,243 - 567 = ?$ Base: $\underline{\hspace{1cm}}$ Complement: $\underline{\hspace{1cm}}$ Answer: $\underline{\hspace{1cm}}$

$9\,612 - 3\,458 = ?$ Base: $\underline{\hspace{1cm}}$ Complement: $\underline{\hspace{1cm}}$ Answer: $\underline{\hspace{1cm}}$

Day 9 — Method Comparison

My heuristic — when do I pick Nikhilam? When do I pick standard borrowing?

Day 10 — Reflection

1. What did you set out to make or learn, and what did you end up with?

2. What is the most surprising thing you discovered?

3. If you had 3 more days, what would you do differently?

Glossary (your own)

Sanskrit	Meaning	First met
<i>Nikhilam Navataścaramam Daśataḥ</i>	“all from 9, last from 10”	Day 1
<i>sūtra</i>	aphorism / formula	Day 1
<i>pūraka</i>	complement	Day 2
<i>vyavakalana</i>	subtraction	(teacher mention)
<i>Tirthaji</i>	author of <i>Vedic Mathematics</i> (1965)	Day 1

Day 1 — The Sūtra and the Hook

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will record a personal baseline time for $10\ 000 - 4567$ using standard borrowing, will recite the Nikhilam sūtra in Sanskrit, and will have watched the method solve the same problem visibly faster.

Materials

- Printed baseline sheet: ONE subtraction problem ($10\ 000 - 4567 = ?$). One per student.
- Stopwatch or wall clock (one per pair is plenty).
- Whiteboard / large paper for the teacher demo.
- A printed Nikhilam sūtra poster — IAST + English — on the wall.

Vocabulary introduced today

- *Nikhilam Navataścaramam Daśataḥ* (IAST: nikhilam navataścaramam daśataḥ; lit. “all from nine and the last from ten”) — the sūtra we’ll learn this fortnight.
- *sūtra* (IAST: sūtra; lit. “thread, formula, aphorism”) — a short mnemonic rule. Tirthaji’s *Vedic Mathematics* gives 16 of them.

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to solve $10\ 000 - 4567$? Make a guess.”
- Take 3–4 guesses. Write the range on the board (e.g. “15 sec to 60 sec”).
- Quick make-10 reflex round: teacher calls digits; class chorus the partner. $8 \rightarrow 2$, $6 \rightarrow 4$, $9 \rightarrow 1$, $3 \rightarrow 7$, $5 \rightarrow 5$, $4 \rightarrow 6$, $7 \rightarrow 3$, $2 \rightarrow 8$.

Baseline timing (8 min)

- Hand out the baseline sheet face-down.
- Students write their name and predicted time on the back.
- On “Go,” flip and solve using whatever method they normally use (standard borrowing).
- Record their actual time (the wall clock is running).
- Mark answer; honest self-correction — they’re racing themselves over the module, not each other.

Meet the sūtra (5 min)

- Point to the wall poster. Teacher reads slowly:
Nikhilam Navataścaramam Daśataḥ “All from nine and the last from ten.”
- Class echoes 3 times: soft / normal / loud.
- Teacher: “That’s the whole rule. Six Sanskrit words. By Day 10 you’ll use it without thinking.”

Teacher demonstration (12 min) — the same baseline problem, the Nikhilam way

Write on the board:

$$\begin{array}{r} 1\ 0\ 0\ 0\ 0 \\ - \quad \quad 4\ 5\ 6\ 7 \\ \hline \end{array}$$

Narrate:

1. “We want to subtract 4567 from 10 000. That’s 10^4 — a clean power of ten. Perfect Nikhilam territory.”
2. “Treat 4567 as a 4-digit number. Apply the rule digit-by-digit.”

Step	Apply rule	Get
First digit (4):	“from 9”	$9 - 4 = 5$
Second digit (5):	“from 9”	$9 - 5 = 4$
Third digit (6):	“from 9”	$9 - 6 = 3$
Last digit (7):	“from 10”	$10 - 7 = 3$

- Read the answer top-to-bottom: **5433**.
- Verify** (with the class): $4567 + 5433 = 10\ 000$. ✓
- “Notice — I never borrowed. Not once. The ‘all from nine’ rule means we’re always subtracting from 9, which is bigger than any single digit.”

Activity (10 min) — First try

- Students re-do the same problem ($10\ 000 - 4567$) on the back of their baseline sheet, using the Nikhilam rule. Refer to the wall poster.
- Time themselves again.
- Compare: how much slower (or faster) was the Nikhilam method on the FIRST try?

Wrap-up (5 min)

- Show of hands: who was faster with Nikhilam on first try? Who was slower? (Most will be slower — that’s expected. Practice fixes it.)
- Preview Day 2: “Tomorrow we drill the foundational skill — finding the complement of a 2-digit number to 100, fast.”
- Set the wall-poster Sanskrit chant as the daily opener for the next 9 days.

Student-facing content

The Nikhilam Sūtra

Nikhilam Navataścaramam Daśataḥ — “all from nine and the last from ten.”

To subtract from 10, 100, 1000, 10 000, ...: 1. Subtract each digit *from 9*, **except the last digit** — subtract it *from 10*. 2. Read the answer top-to-bottom. 3. Verify: original + answer = the round base.

No borrowing. Ever.

Today’s baseline: - My time with my normal method = _____ seconds - My time with Nikhilam (first try) = _____ seconds

I expect the second number to drop by Day 5.

Homework

- Memorise the sūtra. Be able to say it from memory tomorrow.
- Practice: find the complement of these 2-digit numbers to 100:
 - 73, 41, 28, 86, 95, 17, 60, 52, 34, 89.
 - Time yourself. Aim for under 30 seconds for all ten.

Differentiation

- **Tier up:** Try Nikhilam on 100 000 – 23 456. Time it.
- **Tier down:** If the 4-digit problem feels overwhelming, drill complements to 100 (2-digit numbers) for the activity instead.

Teacher notes

- The baseline time is the *single most important* artefact of Day 1. Make sure every student records it honestly. Day 5's retest depends on it.
- Don't promise a specific speedup — say “you'll likely be faster by Day 5; how much depends on you.”
- The Sanskrit pronunciation is hard. Model it slowly; let students laugh on the first attempt. By Day 5 they'll all say it cleanly.
- One paraphrase from Tirthaji that helps: “Nikhilam (निखिलं) means ‘whole,’ ‘entire,’ ‘all.’ Navataḥ means ‘from nine.’ Caramaṁ means ‘the last.’ Daśataḥ means ‘from ten.’ Six words; one method.”

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2). The full sūtra reads *Nikhilam Navataścaramaṁ Daśataḥ*.
- *Microsoft Word - sutras1.pdf* — wall-poster source.

Day 2 — Complement to 100

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will find the complement (*pūraka*) of any 2-digit number to 100 in under 3 seconds, using the Nikhilam rule applied to two digits.

Materials

- Whiteboards or scrap paper for live drills
- A printed wall poster: 9, 9, 9, 9, 9, . . . 10 showing the digit-wise rule visually
- Pair-drill sheets (20 two-digit numbers each)
- Stopwatch (one per pair)

Vocabulary introduced today

- *pūraka* (IAST: pūraka; lit. “that which completes”) — the **complement**: the number you add to fill up to 100, 1000, etc.

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times.
- Make-10 reflex: 10 quick digit-partners.

Core (15 min) — The pattern

1. **Pose:** “What’s the complement of 73 to 100?” Take guesses.
2. Walk through it the Nikhilam way:

$$\begin{array}{r} 100 \\ - 73 \\ \hline \end{array}$$

- First digit (7): $9 - 7 = 2$
 - Last digit (3): $10 - 3 = 7$
 - Answer: **27**.
 - Verify: $73 + 27 = 100$. ✓
3. **Why “all from 9, last from 10”?** Show on the board:
$$99 + 1 = 100$$
 - If we subtract each digit *from* 9 we get the complement to **99**, not 100.
 - The “+1” needed to bridge $99 \rightarrow 100$ is exactly what “the last from 10” provides: $10 - x$ is the same as $(9 - x) + 1$.
 - That’s the whole trick. The last digit absorbs the “+1.” Every other digit just goes “from 9.”

4. **Demo 4 more, quickly, with the class chanting along:**

Number	First digit	Last digit	Complement	Check
41	$9 - 4 = 5$	$10 - 1 = 9$	59	$41 + 59 = 100$ ✓
28	$9 - 2 = 7$	$10 - 8 = 2$	72	$28 + 72 = 100$ ✓
86	$9 - 8 = 1$	$10 - 6 = 4$	14	$86 + 14 = 100$ ✓
95	$9 - 9 = 0$	$10 - 5 = 5$	05 = 5	$95 + 5 = 100$ ✓

5. **Edge case: trailing zero.** What is the complement of 30? Apply the rule: $9 - 3 = 6$ (for tens), $10 - 0 = 10$ (for units). But 10 isn’t a digit — it carries. So you get $6 \mid 10 \rightarrow 70$. Easier: notice that the last *non-zero* digit gets the “from 10” treatment; trailing zeros stay as zeros.

Or more simply: $30 + 70 = 100$, so complement of 30 is 70. The rule still works, you just respect place value.

Activity (20 min) — Paired drill

- Pairs face each other across desks. One holds the drill sheet (20 two-digit numbers); the other says complements out loud.
- Round 1: 60 seconds. Count correct answers.
- Swap roles. Round 2.
- Round 3: swap to a new sheet (different numbers). Aim to beat your Round 1 count.

Wrap-up (5 min)

- Quick poll: who got 15+ in 60 sec? 18+? 20?
- Recall paraphrase: “*To complete to 100, take each digit from 9; the last digit takes from 10.*”
- Preview Day 3: “*Tomorrow — same idea, but complement to 1000.*”

Student-facing content

Complement (*pūraka*) to 100

For any 2-digit number AB:

- Tens digit: $9 - A$
- Units digit: $10 - B$

Examples: - 73 $\rightarrow$ 27 ($9-7=2$, $10-3=7$) - 41 $\rightarrow$ 59 ($9-4=5$, $10-1=9$) - 95 $\rightarrow$ 05 ($9-9=0$, $10-5=5$)

Verify by adding: number + complement = 100.

Tirthaji’s *Vedic Mathematics* (1965) gives the rule as *Nikhilam Navataścaramam Daśataḥ* — and a “complement to 100” is just the rule applied to two digits.

Homework

- Memorise complements to 100 of the multiples of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95. These should become automatic.
- Time yourself: 20 random complements in under 60 seconds. Practise twice.

Differentiation

- **Tier up:** Try complements to 200, 300, 500. The rule modifies — find complement to 100, then adjust. (Tier-up students should be able to articulate that adjustment by the end of class.)
- **Tier down:** Use the rule only when both digits of the number are non-zero. Trailing zeros will be drilled on Day 3.

Teacher notes

- Most common error today: $9 - 0 = 9$ (correct) but $10 - 0 = 10$ (cannot be a digit). Students will write 0 and lose 10. Reinforce: if the last digit is 0, the complement’s last

digit is also 0, and the “10” gets absorbed into the next column. Or: just note that the rule cleanly handles non-zero last digits; trailing zeros are a quick edge-case.

- Many students try to do this in their head WITHOUT going digit-by-digit. That’s fine if they get the right answer — but make sure they can *show* the digit-wise method when asked. Day 6’s “why it works” depends on the digit-wise framing.
- Pair students with different reflexes. The faster-reflex student often models; the slower benefits.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — the source of the digit-wise rule.

Day 3 — Complement to 1000

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will find the complement of any 3-digit number to 1000 in under 4 seconds, and will recognise the same pattern extends to 10 000, 100 000, and beyond.

Materials

- Whiteboard
- Pair-drill sheets (20 three-digit numbers; second set with mixed 3- and 4-digit)
- Stopwatch
- Calculator (for self-verification only)

Vocabulary

- *pūraka* — complement (introduced Day 2; reuse)

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times.
- Quick fire: teacher calls 2-digit numbers; students chorus the complement to 100.
 - 73 → 27, 41 → 59, 28 → 72, 86 → 14, 95 → 05, 60 → 40, 17 → 83.

Core (15 min) — Three digits, same rule

1. **Pose:** “What’s the complement of 738 to 1000?”
2. Walk through:

$$\begin{array}{r} 1000 \\ - 738 \\ \hline \end{array}$$

- First digit (7): $9 - 7 = 2$
 - Second digit (3): $9 - 3 = 6$
 - Last digit (8): $10 - 8 = 2$
 - Answer: **262**.
 - Verify: $738 + 262 = 1000$. ✓
3. “Notice — same six-word rule. We just have one more ‘from 9’ step in the middle.”
4. **The general pattern:**

For an n -digit number, the complement to 10^n is:

$$(9 - d_1), (9 - d_2), \dots, (9 - d_{n-1}), (10 - d_n)$$

where $d_1 \dots d_n$ are the digits left-to-right.

5. **Demo 4 more, with class chanting along:**

Number	Working	Complement	Check
419	9-4, 9-1, 10-9 = 5, 8, 1	581	$419 + 581 = 1000$ ✓
256	9-2, 9-5, 10-6 = 7, 4, 4	744	$256 + 744 = 1000$ ✓
803	9-8, 9-0, 10-3 = 1, 9, 7	197	$803 + 197 = 1000$ ✓
999	9-9, 9-9, 10-9 = 0, 0, 1	001 = 1	$999 + 1 = 1000$ ✓

6. **Quick extension to 10 000:** Complement of 2347 to 10 000?

- 9-2, 9-3, 9-4, 10-7 = **7653**.
- $2347 + 7653 = 10\ 000$. ✓
- “Same rule. Just more digits.”

Activity (15 min) — Speed drill

- Pair-drill: 20 three-digit numbers. Aim for all 20 in under 90 seconds.
- Self-check using a calculator: number + your complement should equal 1000.
- Second sheet: 10 four-digit numbers, complement to 10 000. Aim under 60 seconds for the set.

Wrap-up + Formative quiz (10 min)

- Formative quiz Part A from `quizzes/formative.md` — 5 questions on slips of paper, 5 min.
- Collect. Next class includes the result.

Student-facing content

Complement to 1000 (and beyond)

For any 3-digit number ABC: - First digit: **9 – A** - Second digit: **9 – B** - Last digit: **10 – C**

Same six-word rule. Same approach for 4-digit numbers (one extra “from 9”) and 5-digit numbers (two extra “from 9”s).

Examples: - 738 → 262 - 419 → 581 - 2347 → 7653 (complement to 10 000)

“Nikhilam navataścaramam daśataḥ — *all from nine and the last from ten.*” (Tirthaji, *Vedic Mathematics*, 1965, sūtra #2.) The “all” handles every digit except the last; “the last from ten” handles the last digit’s special role.

Homework

- Find the complement of each of these to 1000:
 - 156, 423, 789, 802, 64 (treat as 064), 7 (treat as 007), 555, 900.
- Find the complement of each of these to 10 000:
 - 3456, 1027, 8888, 75 (treat as 0075), 999.
- Time yourself. Total time goal: under 2 minutes.

Differentiation

- **Tier up:** Complement of 5-digit numbers to 100 000. Time goal: under 6 sec each.
- **Tier down:** Stay on 3-digit complements. The four-digit and five-digit cases come naturally once 3-digit is automatic.

Teacher notes

- The most common error: a number like 64 has only two digits; students forget the rule wants a 3-digit number, so they apply the rule to “64” (giving 36) instead of “064” (giving 936). Make sure they “pad with zeros” mentally.
- Another common error: the last digit being 0. For 800 → complement to 1000: digit-wise gives 9-8, 9-0, 10-0 = 1, 9, 10. The “10” carries. Easier reading: 800 + 200 = 1000, so complement is 200. The rule still works (1, 9, 10 → 1, 10, 0 → 200), but it’s a place where the padding/carry interacts. Use this as a teachable moment; preview Day 6’s algebraic justification.
- Some students will sniff out a shortcut: “the complement just makes everything add to 9 except the last column which adds to 10.” Great — that is the rule, restated. Praise.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).
- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter — worked 3-digit and 4-digit examples.

Day 4 — Subtracting from a Power of Ten

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute subtractions of the form $10^n - A$ (e.g. $1000 - 567$, $10\,000 - 4783$) directly by reading off the complement of A , with no written borrow.

Materials

- Whiteboard
- `activities/activity-01-complement-drill.md` worksheet (printed)
- Stopwatches (one per pair)

Vocabulary

(All terms reused from Days 1–3.)

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: complements to 100, then complements to 1000. Teacher calls; students chorus.

Core (10 min) — The shift from “complement” to “subtraction”

1. **The pivot.** Write on the board:

$$1000 - 567 = ?$$

“We’ve spent two days finding the complement. The complement of 567 to 1000 is 433. Notice that’s also the answer.”

2. **Why?** *“If $A + B = 1000$, then $1000 - A = B$ and $1000 - B = A$. Complement IS the difference, when you’re subtracting from the base.”*
3. **Worked examples, with class predicting:**

Problem	Working	Answer
$1000 - 567$	9-5, 9-6, 10-7 = 4, 3, 3	433
$1000 - 248$	9-2, 9-4, 10-8 = 7, 5, 2	752
$10\,000 - 4567$	9-4, 9-5, 9-6, 10-7 = 5, 4, 3, 3	5433
$10\,000 - 89$	9-0, 9-0, 9-8, 10-9 = 9, 9, 1, 1	9911 (pad 89 → 0089)
$100\,000 - 23\,456$	9-2, 9-3, 9-4, 9-5, 10-6 = 7, 6, 5, 4, 4	76\,544

4. **The padding rule.** *“If A has fewer digits than the base 10^n , pad A with leading zeros. 89 in $10000 - 89$ becomes 0089. The Nikhilam rule needs as many digits as the base has zeros.”*

Activity (20 min) — Complement Drill (timed pairs)

See [activities/activity-01-complement-drill.md](#).

- 8 problems on a sheet. Mixed bases: 4 are $1000 - X$, 3 are $10\ 000 - X$, 1 is $100\ 000 - X$.
- Pair self-correction. Time the pair.

Wrap-up (10 min)

- Compare with standard borrowing: pick one problem (say $10\ 000 - 4567$) and ask half the class to solve by borrowing, half by Nikhilam. Same time limit. See who finishes first.
- Discussion: “*When does Nikhilam beat standard borrowing? When does standard borrowing match it?*” (Answer: Nikhilam dominates when subtracting from a round base; matches roughly when subtracting from a non-round number — that’s the Day 8 topic.)
- Preview Day 5: “*Tomorrow — mid-module retest of the Day 1 baseline. Same problem. Different speed.*”

Student-facing content

Subtracting from a Power of Ten

To compute $10^n - A$:

1. Make sure A has exactly n digits (pad with leading zeros if needed).
2. Apply the Nikhilam rule digit-by-digit: each digit “from 9,” last digit “from 10.”
3. Read off the answer.

No borrowing required. Verify by adding: original + answer = base.

Tirthaji’s sūtra (1965): *Nikhilam Navataścaramam Daśataḥ*. Six words; an entire subtraction shortcut.

Homework

- 12 problems on the back of today’s worksheet. Time yourself. Try to finish all 12 in under 4 minutes.
- One reflection sentence: “*How did the Nikhilam method feel today compared to Day 1?*”

Differentiation

- **Tier up:** Mixed-base problems where the base is given symbolically. E.g. let $B = 10^6 = 1\ 000\ 000$; compute $B - 234\ 567$.
- **Tier down:** Stick to $1000 - A$ for the homework. Step up to $10\ 000$ only when the 1000 case is automatic.

Teacher notes

- This is the day students go “ohh — that’s why we learned the complement.” Make the connection explicit. The complement IS the answer when subtracting from the base.

- A student will inevitably ask: “What if the problem is $9999 - 567$? Does it still work?” Answer: yes, but without the “last from 10” — the answer is $9999 - 567$, computed digit-wise as $9-5, 9-6, 9-7 = 432$. Try it: $567 + 432 = 999$. ✓. So $9999 - 567 = 9432$. This is the special case where ALL digits are “from 9” (because there’s no “+1” needed when the base is $10^n - 1$). Good for Tier-up.
- Avoid letting students treat the method as a black box. Insist they verify at least one of their answers by adding.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).
- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter — provided the 5-digit example pattern.

Day 5 — Mid-module Check

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest the Day 1 baseline problem using the Nikhilam method, record the new time, and complete the formative quiz Part B that consolidates Days 1–4.

Materials

- The Day 1 baseline sheets (returned today)
- Fresh paper for the retest
- `quizzes/formative.md` Part B (printed)
- Stopwatch

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times (loud).
- Make-10 reflex: 10 quick partners.

Retest of Day 1 baseline (8 min)

- Hand back the Day 1 sheets so students can see their old time.
- Give a fresh sheet with the same problem: $10\ 000 - 4567 = ?$
- “Solve it the Nikhilam way. Time yourself.”
- Record new time below the old time.
- % **improvement** = $(\text{Day 1 time} - \text{Day 5 time}) / \text{Day 1 time} \times 100$. Compute on the spot.

Brief sharing (5 min)

- 3 volunteers share their % improvement.

- “Some of you went from 30 sec to 8 sec. Some from 18 sec to 15 sec. Both are wins. The point is the trend.”

Formative quiz Part B (15 min)

- Hand out quizzes/formative.md Part B.
- 12 mixed problems: 3 complement-to-100, 4 complement-to-1000 / $1000 - N$, 3 $1000 - N$, 1 four-digit padding case, 1 short answer (“state the sūtra in English and Sanskrit”).
- Closed notebook. Aim to finish in 12 minutes.
- Collect for grading.

Diagnostic discussion (10 min)

- While the quiz is being graded (or based on a quick teacher scan), call out three common error types on the board:
 1. **Padding error** — student wrote $10000 - 89 = 9921$ because they treated 89 as “8 9” not “0 0 8 9.” Reteach: pad with leading zeros to match the base’s digit count.
 2. **Last-digit error** — student wrote $9 - 7 = 2$ for the units instead of $10 - 7 = 3$. Reteach: the LAST digit always takes from 10. Always.
 3. **Trailing-zero handling** — $1000 - 70$: digit-wise gives 9-0, 9-7, 10-0 = 9, 2, 10. The 10 carries: 9, 3, 0 → 930. Verify: $70 + 930 = 1000$. ✓. Acknowledge this is a corner case; offer the easier “see 70 as 070, complement directly” framing.
- Take questions.

Wrap-up (2 min)

- Project choice form due tomorrow. Three options: shopkeeper chart, teaching video, two’s-complement poster. (See [assessments/project-brief.md](#).)
- Preview Day 6: “Why does this work? Tomorrow we prove it — with algebra.”

Student-facing content

Mid-module Check

Day 1 time: ____ sec → Day 5 time: ____ sec → % change: ____

If you improved less than expected, the most common reason is one of three things:

1. Padding error — you forgot leading zeros.
2. Last-digit error — you used “from 9” for the last digit instead of “from 10.”
3. You’re still translating mentally between “borrow” and “complement.”

Don’t panic. Days 6–9 fix all three.

Homework

- Choose your project format (shopkeeper chart / video / poster) and write a 3-sentence plan.

- 5 mixed Nikhilam subtractions: $1000 - 358$, $10\ 000 - 6427$, $1000 - 72$, $100\ 000 - 12\ 345$, $10\ 000 - 9$. (Last one is 9991; check the padding!)

Differentiation

- **Tier up:** Try mixed-base, mixed-method speed test: 6 problems in 60 seconds. Use Nikhilam where helpful, standard where not.
- **Tier down:** Re-do the Day 4 worksheet. Don't take the quiz cold.

Teacher notes

- The retest is the emotional pivot of the module. Most students will see a real improvement; honor it without overselling. ("You're 50% faster — and you're still getting faster.")
- Use the formative quiz to triage: anyone scoring below 60% gets a 10-min pull-aside on Day 7 before the two's-complement lesson. Don't let them carry a foundational gap into the harder material.
- Distinguish: this is a *formative* check, not summative. No grade weight. Day 10 is summative.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).

Day 6 — Why It Works

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will explain *why* the Nikhilam rule produces the correct answer, using the identity $10^n - A = (10^n - 1) - A + 1$, and will draw a bar-model that shows the same idea visually.

Materials

- Whiteboard
- Worksheet: "explain to a friend" prompt + space
- Coloured pencils for the bar model
- The 16-sūtra list (Microsoft Word - sutras1.pdf) visible

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 4 complements (mix of 2-digit and 3-digit).
- Return graded formatives. Acknowledge the common errors.

Core (20 min) — The mathematical reason

1. The core identity (write boxed on the board):

$$10^n - A = (10^n - 1) - A + 1$$

2. **Why this identity?** “ 10^n is 1 followed by n zeros — e.g. $1000 = 10^3$. $10^n - 1$ is the same number minus one — e.g. 999. That’s just ‘all nines.’ So the identity says: subtracting A from a round base equals (subtracting A from all-nines) plus 1.”
3. **Why does that help?** “Subtracting any digit from 9 is trivial — there’s no borrow. We always have $9 \geq \text{digit}$. So $(10^n - 1) - A$ is a clean digit-by-digit subtraction with no borrow. Then we add 1 at the end.”
4. **Where does the “+1” go?** “Into the last digit. So instead of doing ‘from 9’ on the last digit and then ‘+1,’ we just do ‘from 10’ on the last digit. Same thing. That’s why the sūtra says ‘all from nine and **the last from ten.**’”
5. **Walk through a concrete example with full algebra:**

$$\begin{aligned} 1000 - 567 \\ &= (1000 - 1) - 567 + 1 && \leftarrow \text{identity} \\ &= 999 - 567 + 1 \\ &= (9-5)(9-6)(9-7) + 1 && \leftarrow \text{digit-wise from 9 — no borrow!} \\ &= 4\ 3\ 2 + 1 \\ &= 432 + 1 \\ &= 433. && \checkmark \end{aligned}$$

Now compare with the sūtra direct version:

$$\begin{aligned} 1000 - 567 \\ &= (9-5)(9-6)(10-7) && \leftarrow \text{Nikhilam, last from 10} \\ &= 4\ 3\ 3 \\ &= 433. && \checkmark \end{aligned}$$

“Same answer. The sūtra is just the identity, made into a mnemonic.”

6. Bar model.

Draw a long horizontal bar labelled 1000. Inside, mark off a left segment labelled 567 and a right segment labelled ? (the unknown). “What we want is the right segment.”

Now ask: “What’s the length of the bar minus 1?” — Answer: 999.

Mark a tiny +1 sliver at the right end of the bar. Now $999 = 567 + (\text{right segment} - 1)$, so right segment = $(999 - 567) + 1 = 432 + 1 = 433$.

“That’s the picture-version of the algebra. Same thing.”

Activity (15 min) — Explain to a friend

- Worksheet prompt:

Imagine your Grade 4 cousin asks: “*Why does Nikhilam work? Isn’t it just magic?*” In 4–6 sentences, explain why it’s NOT magic. Use the words **nine**, **ten**, **plus one**, and **place value**.

- 10 min individual writing.
- 5 min pair-share: each partner reads their paragraph; partner says one thing that’s clear and one that’s unclear.

Wrap-up (5 min)

- Read one strong paragraph aloud (with permission).
- Honest historical note: “*Tirthaji’s book (1965) gives this rule. The rule itself is genuinely correct. Tirthaji’s claim that it comes from a Vedic appendix has not been verified by other Sanskritists. We use the rule because it works. The history is younger than the name suggests.*”
- Preview Day 7: “*Tomorrow — a surprise. The same rule is built into how computers subtract. It’s called ‘two’s complement.’*”

Student-facing content

Why Nikhilam works

The key identity:

$$10^n - A = (10^n - 1) - A + 1$$

- $10^n - 1$ is “all nines” (e.g. 9999).
- Subtracting A from “all nines” is trivial digit-by-digit — there’s no borrow because $9 \geq$ any digit.
- The “+ 1” at the end gets absorbed into the last digit, turning “ $9 - \text{last_digit}$ ” into “ $10 - \text{last_digit}$.”

That’s the whole proof.

Tirthaji’s *Vedic Mathematics* (1965) packages this proof into six Sanskrit words: *Nikhilam Navataścaramam Daśataḥ* — “all from nine and the last from ten.” The math is real. The Sanskrit is from a 20th-century synthesis, not a Vedic text.

It is not magic. It is good notation.

Homework

- Find ONE moment in your daily life where you naturally “round to the nearest 100 / 1000” without thinking. Write 2 sentences describing it. (E.g. tipping at a restaurant. Counting change. Comparing prices.)
- Practise: complement of 4321 to 10 000. Show the algebraic identity for this one (the same way the lesson did $1000 - 567$).

Differentiation

- **Tier up:** Write the explanation for a Grade 11 student. Use formal notation. Show the identity holds for any base b : $b^n - A = (b^n - 1) - A + 1$. Preview Day 7's binary case.
- **Tier down:** Skip the written explanation. Verbalise to your partner. Use the bar-model picture as scaffold.

Teacher notes

- The most common student error in the paragraph: confusing “the last digit takes from 10” with “you add 1 at the end.” These are the SAME operation, just expressed differently — $10 - x = (9 - x) + 1$. Some students will write both; that's fine.
- Some students prefer the visual bar-model; some prefer the algebra. Both are valid. Honour the choice and ask them to draw the OTHER one as homework.
- Honest provenance is essential here. Don't oversell. Some students (or parents) believe Nikhilam is “ancient Vedic mathematics.” Clarify gently; preview the Senior-band treatment of the historicity question.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.
- *Microsoft Word - sutras1.pdf* — wall reference for the full 16 sūtra list.

Day 7 — Two's Complement Connection

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will identify the structural similarity between Nikhilam (“all from 9, last from 10”) in base 10 and **two's complement** (“flip all bits, add 1”) in base 2, and will subtract two small binary numbers using the two's-complement method.

Materials

- Whiteboard
- Printed 8-bit binary number line poster (0000 0000 to 1111 1111)
- Pencil + ruled paper
- Optional: a calculator with a binary mode (most smartphone calculators have one)

Vocabulary introduced today

- *binary* — base 2 (the number system computers use).
- *bit* — a binary digit; either 0 or 1.
- *two's complement* — the binary version of the Nikhilam principle.

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 3 base-10 complements (to 100, 1000, 10 000).
- Provocation: “*Same trick. Different number system. Where else might this work?*”

Core (25 min) — From base 10 to base 2

1. The Nikhilam pattern, restated in base-neutral language:

To compute $b^n - A$:

- Each digit of A : subtract from $(b - 1)$ — the maximum digit in that base.
- The last digit: subtract from b .

In base 10: $(b - 1) = 9$ and $b = 10 \rightarrow$ “all from 9, last from 10.”

2. In base 2. “*The biggest digit in base 2 is 1. So ‘(b - 1)’ is 1. The base b is 2.*”

Pattern becomes:

- Each bit of A : subtract from 1. (This is the same as **flipping the bit**: $0 \rightarrow 1, 1 \rightarrow 0$.)
- The last bit: subtract from 2. ($2 - 0 = 10$ in binary, $2 - 1 = 1$. Equivalent: “flip the last bit and add 1” — except we just add 1 to the whole flipped result and let the carry propagate.)

That’s exactly the **two’s complement** algorithm taught in computer science: > **To negate a binary number: flip all bits, then add 1.**

3. Worked example. Subtract $5 - 3$ using two’s complement, 4 bits:

- $5 = 0101$. $3 = 0011$.
- To compute $5 - 3$, we compute $5 + (-3)$, where -3 is the two’s complement of 3.
- Two’s complement of 3 (4 bits): flip all bits of $0011 \rightarrow 1100$, then add 1 $\rightarrow 1101$.
- Now add: $0101 + 1101 = 1\ 0010$.
- The leading 1 (the 5th bit) is the **overflow / carry out** — we discard it. (This carry-out is the “underflow” of the $16 = 2^4$ base, exactly like the $10\ 000$ in $10\ 000 - 4567$.)
- Result: $0010 = 2$. ✓

4. Why this is the SAME as Nikhilam.

- In base 10: $10\ 000 - 4567 = 5433$. Same calculation: complement of 4567 = 5433, and $4567 + 5433 = 10\ 000$.
- In base 2: $16 - 3 = 13 = 1101$. That’s the two’s complement of 3. And $0011 + 1101 = 1\ 0000 = 16$. ✓

- Then we subtract THAT result from 5 (or equivalently add it to 5 modulo 16): $5 - 3 \equiv 5 + (16 - 3) \pmod{16} = 5 + 13 = 18 \equiv 2 \pmod{16}$.

“Tirthaji’s six-word sūtra and the two’s-complement circuit in your phone’s CPU are doing the same algebra. The CPU just does it in base 2.”

5. **Why computers use this.** A CPU has no separate “subtract” circuit — it has an “add” circuit, an “invert bits” circuit, and a “+1” (increment) circuit. Two’s complement lets it do subtraction with just those. This is the deepest reason: the Nikhilam principle saves silicon.

Activity (10 min) — Try it

- Worksheet: 4 small subtractions to do via two’s complement (4 bits, all results positive).

Problem	A	B	Flip A or B?	Two’s-comp	Add	Discard top bit	Answer
6 – 4	0	0	Flip B	1100	1	drop top → 0010	2 ✓
	1	1			001		
	1	0			0		
	0	0					
7 – 5	0	0	Flip B	1011	1	drop top → 0010	2 ✓
	1	1			001		
	1	0			0		
	1	1					
9 – 2	1	0	Flip B	1110	1	drop top → 0111	7 ✓
	0	0			011		
	0	1			1		
	1	0					
12 – 7	1	0	Flip B	1001	1	drop top → 0101	5 ✓
	1	1			010		
	0	1			1		
	0	1					

- Students work in pairs. One does the binary math; the other verifies with a calculator.

Wrap-up (5 min)

- “You’ve just done in your head what a CPU does in nanoseconds. Different scale, same logic.”
- Preview Day 8: “Tomorrow — what if the number we’re subtracting FROM isn’t a clean power of ten? E.g. $8243 - 567$. Nikhilam still helps, but we need one more move.”

Student-facing content

Two’s Complement = Nikhilam in Binary

	Base 10 (Nikhilam)	Base 2 (Two’s Complement)
Maximum digit	9	1
“All from (max)”	All from 9	Flip each bit

	Base 10 (Nikhilam)	Base 2 (Two's Complement)
	“Last from base”	Add 1
	Last from 10	
	Same algebra. Same trick. The CPU inside your phone uses this every time it computes a subtraction.	
	“ <i>Nikhilam Navataścaramam Daśataḥ</i> ” — Tirthaji’s 1965 sūtra. The base-2 cousin was independently discovered in 1945 by the team designing the EDVAC computer.	

Homework

- Convert $13 - 5$ to binary (4 bits) and solve via two’s complement. Show your work.
- One reflection sentence: “*What surprised me about the binary connection.*”

Differentiation

- **Tier up:** Try **negative results**. $3 - 5$ in 4 bits gives 1110, which is the two’s-complement representation of -2 . Look up how CPUs read this.
- **Tier down:** Skip the binary math. Just understand the structural parallel: “flip the digits, add 1, same idea.” Repeat the base-10 examples instead.

Teacher notes

- This is the most cross-disciplinary day of the module. Some students will be enthralled; others will find binary intimidating. Both reactions are fine.
- Many students have *heard* of binary; few have *done* arithmetic in it. The 4-bit examples are deliberately small.
- A natural question: “*Does this mean Tirthaji invented two’s complement?*” — No. Two’s complement was independently developed for digital computing in the 1940s. What’s beautiful is the **structural parallel** — the same algebraic principle $(b^n - A = (b^n - 1) - A + 1)$ plays the same role in both base 10 and base 2.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).
- Patterson & Hennessy, *Computer Organization and Design*, two’s-complement section — standard CS reference for the binary parallel.

Day 8 — General Subtraction with Complements

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will subtract a number from any larger number (not necessarily a power of ten) using a *combined* method: take the complement of the subtrahend to a nearby base, add it to the minuend, and adjust by the difference.

Materials

- Whiteboard
- activities/activity-02-shopkeeper.md worksheet
- Pretend currency (paper notes of 100, 500, 1000 rupee denominations)
- Stopwatches

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 4 complements (mix of bases).

Core (15 min) — Beyond clean bases

1. **The setup.** “So far we’ve done $1000 - 567$. Easy: complement of 567 is the answer, 433. Now: $8243 - 567$. The minuend isn’t a clean power of ten. What do we do?”

2. **Idea: borrow a base.** Rewrite:

$$\begin{aligned} &8243 - 567 \\ &= 8243 - 1000 + (1000 - 567) \\ &= 7243 + 433 \\ &= 7676. \end{aligned}$$

“We pick a base that’s larger than the subtrahend ($1000 > 567$ here). We subtract that base from the minuend ($8243 - 1000 = 7243$; easy). Then we add the complement (433). Done.”

Verify: $8243 - 567 = 7676$ by standard borrowing. ✓

3. **Why does this work?** Algebra:

$$M - S = M - (B - (B - S)) = M - B + \text{complement}(S \text{ to } B)$$

“ $B - S$ is the complement of S to base B . Add the complement and subtract B from M .”

4. **Worked examples (class participates):**

Problem	Base B	Complement	$M - B$	Add	Answer
$8243 - 567$	1000	433	7243	$7243 + 433$	7676
$5172 - 89$	100	11	5072	$5072 + 11$	5083
$6300 - 248$	1000	752	5300	$5300 + 752$	6052
$12\,000 - 4\,567$	10\,000	5\,433	2\,000	$2\,000 + 5\,433$	7\,433

5. **When is this faster than standard borrowing?** “When the subtrahend is close to a base — say 567 to 1000 — Nikhila-with-adjustment is genuinely faster. When the subtrahend

is in the middle (like 4567 from 8243, where 4567 is not close to any clean base), standard borrowing is competitive. The mature mental-arithmetic move is to know both and pick.”

Activity (20 min) — Shopkeeper role-play

See `activities/activity-02-shopkeeper.md`.

- In pairs, one is the shopkeeper, one is the customer.
- Customer hands over a “round” note (100, 500, 1000, or 2000 rupees).
- Shopkeeper computes the change using Nikhilam (mental).
- Swap roles every 3 rounds.
- 10 scenarios per pair. Times recorded.

Wrap-up (5 min)

- Quick discussion: “*Where in real life does this method save time?*” Most answers will be shop-related. Affirm and add: change-counting, expense splitting, deficit tracking.
- Preview Day 9: “*Tomorrow — speed test, both methods, head-to-head.*”

Student-facing content

General Subtraction with Complements

To compute $M - S$:

1. Pick a base B such that $B > S$ and B is a power of 10 (so 100, 1000, 10 000, ...).
2. Compute the complement of S to B (using Nikhilam) — call it C .
3. Compute $M - B$ (which is easy, since B is a round number).
4. Answer = $(M - B) + C$.

Algebraic identity: $M - S = (M - B) + (B - S)$

Verify by checking: $M - S$ should equal your answer.

Example: $8243 - 567 = (8243 - 1000) + 433 = 7243 + 433 = \mathbf{7676}$.

Homework

- Solve 6 mixed subtractions using the combined method:
 - $7234 - 587$ (base 1000)
 - $9\,612 - 3\,458$ (base 10 000)
 - $5\,100 - 67$ (base 100)
 - $13\,405 - 2\,850$ (base 10 000)
 - $800 - 39$ (base 100)
 - $25\,000 - 4\,999$ (base 10 000)
- For each, show the base, the complement, and the final answer.

Differentiation

- **Tier up:** What if the subtrahend has *more* digits than the base needs? E.g. $8243 - 5567$ — using base 10 000, complement of 5567 is 4433, and $8243 - 10\ 000 = -1757$, so answer $= -1757 + 4433 = 2676$. Verify: $8243 - 5567 = 2676$. ✓. Negative intermediate results are fine.
- **Tier down:** Use base 100 for everything. Only graduate to 1000 when comfortable.

Teacher notes

- The combined method introduces a real subtlety: students must now choose a base. Some over-pick (use 10 000 for $5172 - 89$ when 100 would do). Some under-pick (use 100 for $8243 - 567$). Either causes errors. Teach: “pick the smallest base that still exceeds the subtrahend.”
- The shopkeeper role-play tends to make this concrete. Make sure the “round notes” are physically present — the mental model “I have 1000 in my hand, I need to give 567 of it” makes the algebra intuitive.
- Some classes will benefit from spending the whole lesson on the shopkeeper drill and pushing the algebra to Day 9. That’s a fine reordering.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.
- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter — provided the general-case worked examples that informed today’s table.

Day 9 — Speed Practice + Method Comparison

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve a mixed set of subtraction problems using both standard borrowing and Nikhilam, will compare the two methods on each problem, and will articulate a personal heuristic for when to pick which method.

Materials

- Comparison worksheet (10 problems, two-column answer space — one column per method)
- Stopwatches
- Whiteboard
- Final project work-time at the end

Lesson flow

Warm-up (5 min)

- Sūtra chant.

- Quick fire: 4 complements (mixed bases).
- Acknowledge: tomorrow is the summative.

Speed test (15 min) — head-to-head

Distribute the comparison worksheet. 10 problems, ordered roughly:

#	Problem	Likely faster method	Why
1	100 – 37	Nikhilam	Pure power-of-ten subtraction
2	87 – 24	Standard	Small, no borrow either way
3	1000 – 463	Nikhilam	Power-of-ten
4	532 – 178	Standard	No clean base nearby
5	10 000 – 5827	Nikhilam	Power-of-ten
6	7000 – 248	Nikhilam (with adjust)	Round multiple of 1000
7	9385 – 4716	Tie / standard	Mid-range subtrahend
8	50 000 – 12 345	Nikhilam (with adjust)	Round multiple of 10 000
9	8243 – 567	Nikhilam (with adjust)	Subtrahend close to 1000
10	4321 – 4118	Standard	Both small, similar magnitudes

Students solve each problem **once** by their chosen method, then again by the other method. Time both. Mark which method was faster.

Discussion (10 min) — When does each win?

Compile on board: which problems were faster with Nikhilam? Which with standard?

Heuristic that students will surface:

- **Nikhilam wins** when the minuend is a clean power of ten OR the subtrahend is close to a clean power of ten.
- **Standard wins** when both numbers are mid-range and the subtraction requires only one or two borrows.
- **Tie** when the numbers are small.

Write the heuristic on the board. Encourage students to copy it into their workbook.

Honest note: *“This is a real piece of mathematical maturity — knowing when to use which tool. Speed comes from knowing the choice, not from forcing one method everywhere.”*

Project work (12 min)

- Last in-class time before the showcase. Pairs work on their project artefact.
- Teacher circulates. Check: does each project have at least one Nikhilam-method example? Does each cite a source?

Wrap-up (3 min)

- Tomorrow’s plan: 20-min summative quiz, then 3-min project presentations.
- Sūtra chant — final practice for tomorrow’s recitation.

Student-facing content

When to use Nikhilam vs. Standard Borrowing

Situation	Faster method
Subtracting from 100, 1000, 10 000, ...	Nikhilam
Subtracting a small number from a round multiple of 100/1000	Nikhilam with adjust
Both numbers in the middle, similar sizes	Standard
Both numbers small (under 100)	Either

The mature move: know both methods; pick the right one for the problem.

Homework

- Final project touches.
- 5 minutes of sūtra recitation practice — say it 5 times before bed.
- Bring all project materials tomorrow.

Differentiation

- **Tier up:** Mental challenge — do 10 mixed problems in your head with no paper. Time goal: under 3 minutes.
- **Tier down:** Worksheet only. Don't worry about choosing methods — solve each by Nikhilam where you can, standard otherwise.

Teacher notes

- The honest framing of “both methods are tools” matters. Avoid letting students walk away thinking “Nikhilam is always faster.” It isn't. It's faster for *certain shapes* of problem.
- Some students will discover the heuristic naturally. Highlight their wording — let them name the rule.
- Project check is critical today. If any pair is behind, give them homework instructions explicitly.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.

Day 10 — Final Assessment + Project Share

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will complete the summative quiz, present their project artefact in 3 minutes, and submit a brief written self-reflection.

Materials

- `quizzes/summative.md` (printed, one per student)
- Stopwatch for project presentations
- Camera (phone) for documenting projects, if permitted by school policy
- Stickers for completion (optional, but motivating)

Lesson flow

Warm-up (3 min)

- Final sūtra recitation, in unison, from memory — *Nikhilam Navataścaramaṁ Daśataḥ*.
- One sentence each, going around the room: “*The one thing I'll remember from this module is...*”

Summative quiz (20 min)

- Closed notebook.

- 25 marks. Sections: complement recall, subtraction practice, general subtraction, conceptual short answer, open response on “why it works.” See [quizzes/summative.md](#).
- Strict 15-min time limit on the multiple-choice / short-answer parts; remaining time for the open response.
- Collect.

Project presentations (18 min)

- Each pair presents for 3 minutes (set the timer visibly).
- Format depends on artefact:
 - **Shopkeeper chart:** Pair demos using pretend currency on a real scenario.
 - **Teaching video:** Pair plays the 60-sec video, then takes 90 sec of Q&A.
 - **Two’s-complement poster:** Pair walks through the poster’s argument.
- Rest of class fills in a quick “feedback card” — one strength, one question.

Wrap-up (4 min)

- Quick poll: was Module 9 useful? Surprising? Hard?
- Honest history note one more time: *“You learned a real, useful method. The sūtra is from Tirthaji’s 1965 book. The deeper history is more recent than the name ‘Vedic’ suggests. You can hold both: the method is real; the historical claim is unverified.”*
- Self-reflection slip (1 minute): on the back of the quiz, complete:
 - One thing I learned: _____
 - One thing I’m still confused about: _____
 - My favourite Sanskrit term from the module: _____

Student-facing content

Today’s tasks

1. Summative quiz (20 min, closed notebook).
2. Project presentation (3 min, with your partner).
3. Self-reflection (1 min, on the back of the quiz).

You’ve come a long way. Day 1’s $10\ 000 - 4567$ took most of you about 30 seconds. By now you can do it in 5.

Homework

- None. Module ends.

Differentiation

- **Tier up:** If finished early on the quiz, write the algebraic identity for the Nikhilam method in full generality (for any base b).
- **Tier down:** Extra 5 min on the quiz allowed for slow processors; quiet pull-aside support.

Teacher notes

- The summative is the most marked single assessment of the module. Be strict on closed-notebook BUT generous on the open-response question — student paraphrases of the algebraic identity are valid even if not perfectly phrased.
- Project presentations: keep the energy up. Use stickers / acknowledgment for every pair. The presentation is partly a performance; affirm the work.
- If a pair didn't finish their artefact, still let them present. The presentation itself is graded; the artefact is graded separately on the rubric.
- File the quiz results AND the project rubric scores to inform pacing for next year's cohort.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).

Writing Prompts

Writing Prompts — Module 9 Nikhilam Subtraction (Middle)

Eight prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

- 1. (Day 1, end of class, 3 min)** Describe what it felt like to subtract $10\ 000 - 4567$ under time pressure. Did your mind borrow? Did you count? Did you panic? Be honest — there's no wrong answer.
- 2. (Day 5, after the retest, 5 min)** You re-solved the same problem with Nikhilam. Did your time change? Did the *feeling* of doing the math change? Compare the two experiences in your own words.

Explanatory prompts

- 3. (Day 3 or Day 6, homework, ~120 words)** Explain Nikhilam to a Grade 4 student who has never seen it. No more than 6 sentences. Use one specific example with verification.
- 4. (Day 7, homework, ~120 words)** Explain in your own words why the same idea works in base 2 (binary) as in base 10. Use one binary example. Don't use the phrase "two's complement" without explaining what it means.

Argumentative prompt

- 5. (Day 6 or homework, 1 paragraph)** Some people claim the Nikhilam sūtra is "ancient Vedic mathematics." Others say it's a 20th-century synthesis published by Tirthaji in 1965. Pick a position and defend it in 5–7 sentences. Reference what you know.

Creative prompts

6. (Day 4 or Day 5, optional, 1 page) Write a short script (50–150 words) in which a frustrated student and a confident Nikhilam user have a conversation. The frustrated student raises one good objection to the method. The confident user gives an honest answer. (No “and then they magically agreed” endings.)

7. (Day 7 or Day 8, optional, 1 page) Imagine the CPU inside your phone could talk. It explains how it just subtracted $7 - 3$. Write a 100–200 word monologue from the CPU’s point of view, in everyday language, but accurate about what it actually does.

Cross-disciplinary prompt

8. (Day 7, optional, 1 paragraph) This module touched mathematics, computer science, and Sanskrit. In 3–5 sentences, describe one *other* place in your life or studies where the same idea (rounding up to a base, adjusting by the difference) shows up.

Prompt-to-band notes

- For **Primary** band, prompts 1, 3, and 6 work as drawings or oral responses.
- For **Senior** band, prompt 5 deserves a full essay treatment with citations from sources.md. Prompt 4 can be extended into a base- n general analysis.

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction. Listening, not judging.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the rubric — does the writing acknowledge the other side, give a specific example, reach a clear position?
- Prompts 6, 7, 8: no grading. Read the best ones aloud (with permission).

Homework

Homework — Module 9 Nikhilam Subtraction (Middle)

Day 1 → Day 2

Task: Memorise the sūtra. Practice 10 complements to 100: $73 \rightarrow ?$, $41 \rightarrow ?$, $28 \rightarrow ?$, $86 \rightarrow ?$, $95 \rightarrow ?$, $17 \rightarrow ?$, $60 \rightarrow ?$, $52 \rightarrow ?$, $34 \rightarrow ?$, $89 \rightarrow ?$ Time goal: under 30 seconds total.

Time: 5 min.

Day 2 → Day 3

Task: Have a parent or sibling call out 20 two-digit numbers in 90 seconds; you say the complement to 100 for each. Track how many you got right.

Time: 5 min.

Day 3 → Day 4

Task: Find the complement to 1000 of each: 156, 423, 789, 802, 64 (→ 064), 7 (→ 007), 555, 900. Find the complement to 10 000 of each: 3 456, 1 027, 8 888, 75 (→ 0075), 999. Total time goal: under 2 minutes.

Time: 10 min.

Day 4 → Day 5

Task: 12 power-of-ten subtractions on the back of today's worksheet. Self-check by adding (number + your answer should equal the base).

Time: 15 min.

Day 5 → Day 6

Task: Submit your project choice + a 3-sentence plan. (One of: shopkeeper chart / teaching video / two's-complement poster.) Plus: 5 mixed subtractions — $1000 - 358$, $10\ 000 - 6\ 427$, $1000 - 72$, $100\ 000 - 12\ 345$, $10\ 000 - 9$.

Time: 15 min.

Day 6 → Day 7

Task: Find ONE moment in your daily life where you naturally “round to the nearest 100 / 1000” without thinking. Write 2 sentences. Example: rounding a restaurant tip, counting change, comparing prices.

Plus: redo today's algebra example with $10\ 000 - 4321$. Show the identity $10^n - A = (10^n - 1) - A + 1$ for this case.

Time: 10 min.

Day 7 → Day 8

Task: Convert $13 - 5$ to binary (4 bits) and solve via two's complement. Show your work. Plus: one reflection sentence — what surprised you about the binary connection.

Time: 10 min.

Day 8 → Day 9

Task: 6 mixed general subtractions: $- 7\,234 - 587$ (base 1000) - $9\,612 - 3\,458$ (base 10 000) - $5\,100 - 67$ (base 100) - $13\,405 - 2\,850$ (base 10 000) - $800 - 39$ (base 100) - $25\,000 - 4\,999$ (base 10 000)

For each, show your chosen base, the complement, and the answer.

Time: 20 min.

Day 9 → Day 10

Task: Final project touches. 5 minutes of sūtra recitation. Bring all project materials. Be ready to present in 3 minutes.

Time: 15 min plus material gathering.